Introduction to Aeronautics

Lecture | Prof. Dr.-Ing. Mirko Hornung

Academic Year 2026



Contact

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Agenda (1)

1. History of Civil Air Transport

1.1 Milestones of Human Flight

1.2 Major Technological Prerequisites

1.3 Commercialization of Air Transport

2. Introduction to Modern Civil Air Transport

2.1 Facts and Figures

2.2 Market Outlook and Fleet Development

2.3 Future Challenges in Aviation

2.4 Major Players

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2.6 Safety

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3.4 Aircraft Cabin Design

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4.1 Business Models and Networks

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5. Airport Operations

5.1 Airport Elements and Operations

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Agenda (2)

6. Commercial Aircraft Operations

6.1 Regulations and Scheduling

6.2 The Flight Deck

6.3 Flight Crew Activities during a Typical Flight

7. Environmental Impact of Aviation

7.1 Aircraft Noise

7.2 Aircraft Exhaust Emissions

8. Urban Air Mobility

8.1 History

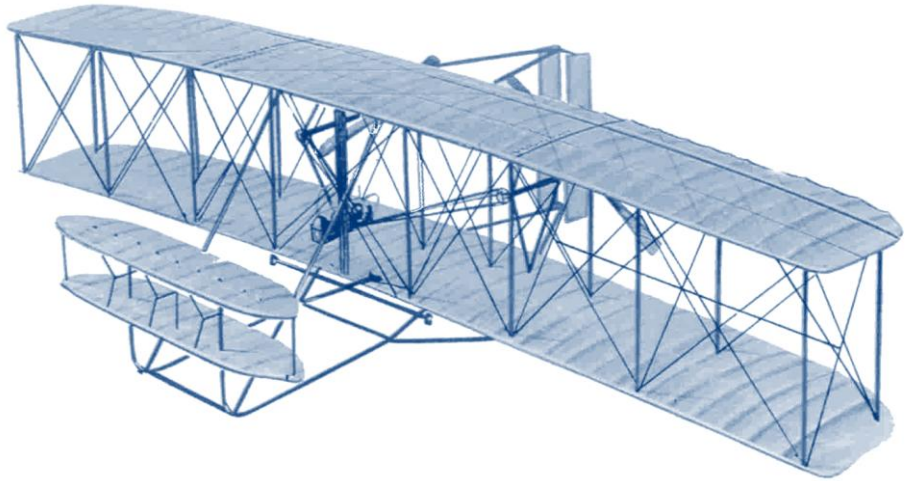
8.2 Use Cases, Potential Benefits, Obstacles

8.3 Vehicle Concepts

8.4 Concept of Operations

8.5 Conclusion

1. History of Civil Air Transport



1. History of Civil Air Transport

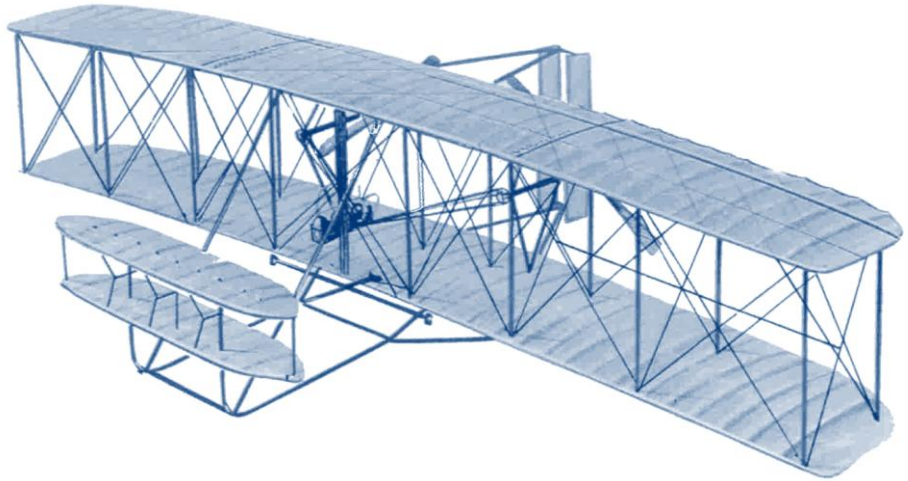
1.1 Milestones of Human Flight

1.2 Major Technological Prerequisites

1.3 Commercialization of Air Transport

1. History of Civil Air Transport

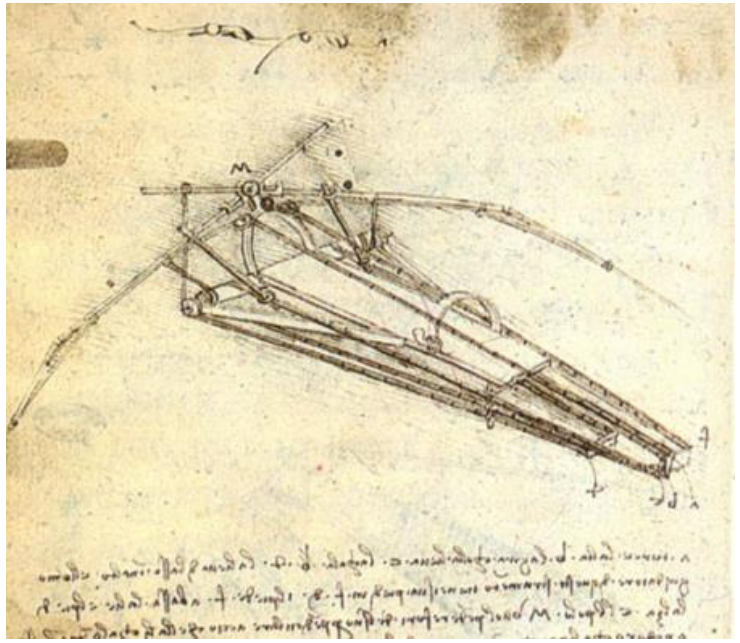
1.1 Milestones of Human Flight



Milestones of Human Flight

1485: Leonardo da Vinci studied the flight of birds and designed an **ornithopter flying machine** (that was never built though).

1783: Montgolfier Brothers demonstrate flight capability of a **hot-air balloon (Lighter-than-Air Vessel)**.





Sources: Institut de France, United States Library of Congress

Milestones of Human Flight: Heavier-than-air Vessels

Sir George Cayley (1773-1857) discovered the principles of manned flight.

- Generation of **lift** required.
- Generation of **thrust** required.
- Air vehicle needs to be **dynamically stable and controllable**.
- Using **Light-weight** construction.
- Separating the forces of the air into **lift** and **drag**.
- Independent **optimization** of air vehicle subsystems.



Source: National Portrait Gallery, London

Milestones of Human Flight: Heavier-than-air Vessels

1894: “Normalsegelapparat” developed by German engineer **Otto Lilienthal** in Berlin.



Source: Otto-Lilienthal-Museum

Milestones of Human Flight: Heavier-than-air Vessels

1903: First **sustained flight** of a heavier-than-air aircraft carrying an adult human: the Wright-Flyer.



Source: Library of the Congress

Milestones of Human Flight: Heavier-than-air Vessels

1927: Solo flight with the *Spirit of St. Louis* (Ryan M-2) from **New York to Paris** in 33 hours 29 min (Charles Lindbergh).



Source: Missouri History Museum

Key features:

- Large fuel tank
- High-efficiency engine (SFS + reliability)
- Single wing with thick airfoil
- High lift-to-drag ratio
- Proved feasibility of world-wide passenger flights

Milestones of Human Flight: Heavier-than-air Vessels

1935: First flight of the Douglas DC-3 – an aircraft **extremely popular** with airlines; also used for military transport purposes.



Source: planespotters.net

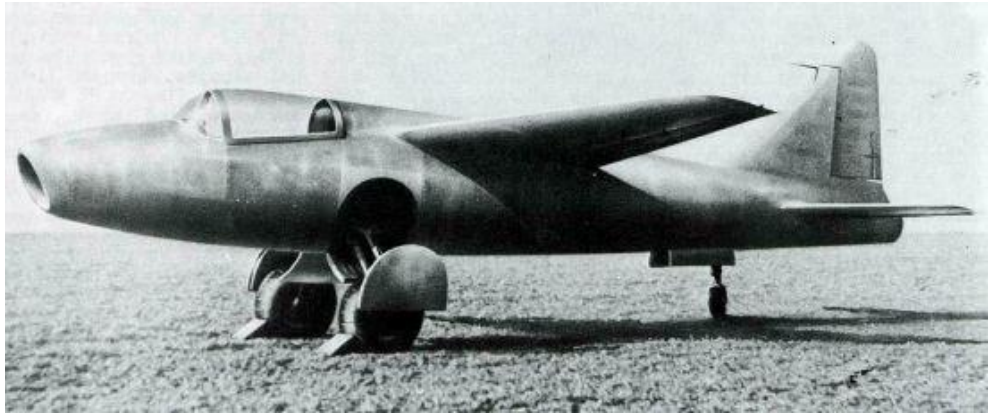
Key features:

- Twin-engine all-metal air transport
- 2x 1200-hp Pratt & Whitney engine
- 28 passengers
- 2,000 miles @ 200 mph

Milestones of Human Flight: Heavier-than-air Vessels

1939: First flight of the Heinkel He 178 – the first aircraft being propelled by a **turbojet engine**.

1942: First flight of the Messerschmitt Me 262 – the first operational **jet-powered** fighter aircraft.



Key features He 178:

- One jet engine delivering 4,4-kN of thrust
- Max. speed 580 km/h at MSL
- Wing span 7.10 m
- Length 7.51 m
- Weight 1.590 kg

Source: historywarsweapons.com

Milestones of Human Flight: Heavier-than-air Vessels

1949 / 1957: First flights of the De Havilland Comet and the Boeing 707 the first production **commercial jet airplanes**.

"…literally changed the world. Travel between far-distant capitals and trade centers has become commonplace." (Brandt, 2004)





Source: aerospaceweb.org, planespotters.net

Milestones of Human Flight: Heavier-than-air Vessels

1969: First flight of the Boeing 747 – the first **Widebody** passenger aircraft.



Source: Alexander Nieder – planespotters.net

Milestones of Human Flight: Heavier-than-air Vessels

1969: First flight of the Aérospatiale-BAC Concorde – one of two commercial **Supersonic Transport** aircraft types ever built.



Source: Museum of Flight

Milestones of Human Flight: Heavier-than-air Vessels

1967 / 1987: First flights of the Boeing 737 and the Airbus A320, the **best-selling commercial aircraft** ever built.



Total Orders: 19,786 units*

Total Deliveries: 11,965 units*



Total Orders: 20,303 units*

Total Deliveries : 12,708 units*

Currently in operation: 11,726

Source: planespotters.net, 737.org.uk, airbus.com

**as of September 2025*

Milestones of Human Flight: Heavier-than-air Vessels

2005: First flights of the Airbus A380, the **largest civil transport aircraft** ever built.



Key features:

- 853 passengers (max.)
- 15,700 km range
- 80 m wing span
- Max. take-off weight: 560 tons
- Max. operating Mach number: 0.89

Source: airbus.com, planespotters.net

Milestones of Human Flight: Heavier-than-air Vessels

2009 / 2013: First flights of the Boeing 787 and the Airbus A350, next generation commercial airliners utilizing ~50% **composite materials**.





Boeing 787-8:
- 242 passengers
- 14,500 km range
- 60 meters wing span

Airbus A350-900:
- 315 passengers
- 14,350 km range
- 65 meters wing span

Source: boeing.com, airbus.com, planespotters.net

Milestones of Human Flight: Heavier-than-air Vessels

2013: First flight of the Bombardier C-Series (A220), the first commercial airliner to use a **geared-turbofan** propulsion system.



Key features (CS300):
- 160 passengers (max.)
- 5,500 km range
- 35 m wing span
- Max. take-off weight: 65 tons
- Max. cruising Mach number: 0.82

Source: bombardier.com

Milestones of Human Flight: Heavier-than-air Vessels

25 September 2014: First flight of the Airbus A320neo (**new engine option**)

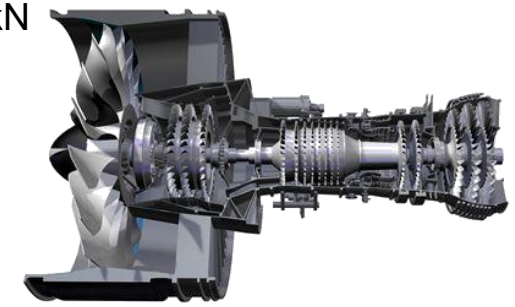


Key features:
- 150 – 180 Pax
- MTOW: 79 t / Range: 6,850 km
- Entry into service: Q4 2015
- Incorporating sharklets and 2 new engine options
- won over 3,000 orders from 65 customers

CFMI LEAP-1A
Max. static thrust: 146 kN
Stages: 1-3-10-2-7
Bypass ratio: 11:1
OPR: 40:1



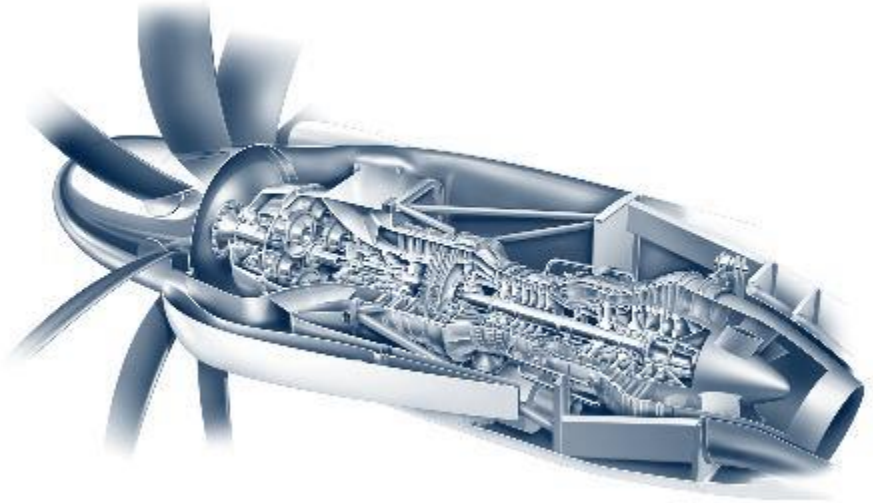
P&W PurePower PW1127G-JM
Max. static thrust: 119 kN
Stages: 1-G-3-8-2-3
Bypass ratio: 12.5:1
OPR: “almost 50:1”



Sources: Airbus, Pratt & Whitney, CFMI, Aviation Week

1. History of Civil Air Transport

1.2 Major Technological Prerequisites



Major Technological Prerequisites

Technological Progress in the following areas has enabled today's practice in civil air transport:

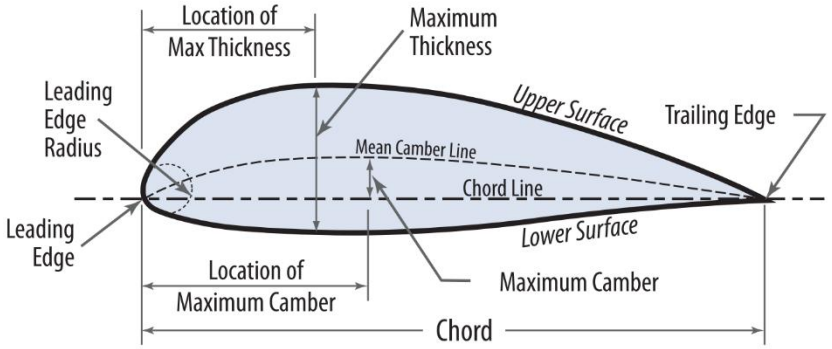


- Navigation
- Generation of Air Lift
- Propulsion
- Automation
- Light-Weight Construction

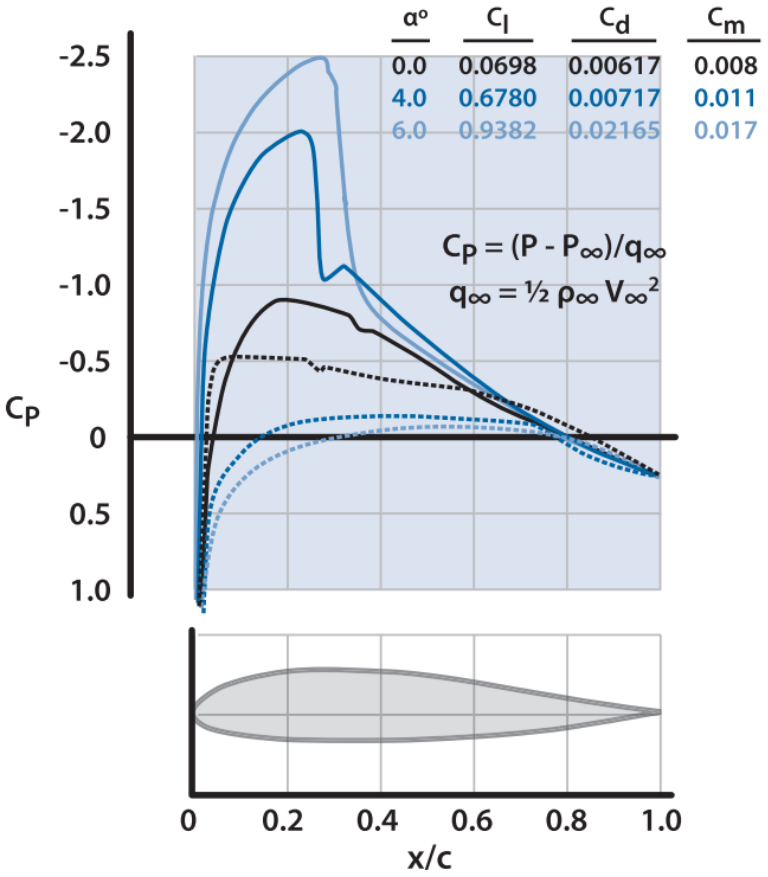
Generation of Air Lift (1)

Airfoil Design

The shape of the airfoil induces a **pressure difference** between the upper and lower surface of the wing.



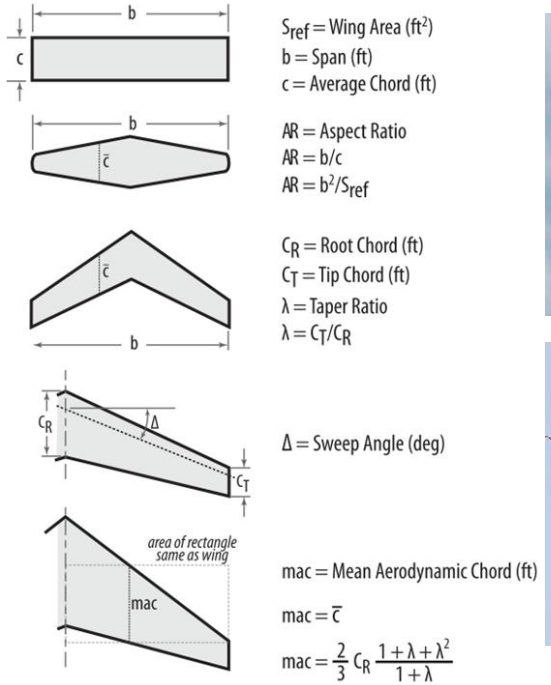
Source: Nicolai (2010)



Generation of Air Lift (2)

Wing Design

Various design parameters define the performance characteristics of the wing.



S_{ref} = Wing Area (ft^2)
b = Span (ft)
c = Average Chord (ft)

AR = Aspect Ratio
$AR = b/c$
$AR = b^2/S_{ref}$

C_R = Root Chord (ft)
C_T = Tip Chord (ft)
λ = Taper Ratio
$\lambda = C_T/C_R$

Δ = Sweep Angle (deg)

mac = Mean Aerodynamic Chord (ft)

$mac = \bar{c}$

$mac = \frac{2}{3} C_R \frac{1+\lambda+\lambda^2}{1+\lambda}$

Sailplane



Cessna 172



Airbus A330





Panavia Tornado



Eurofighter Typhoon



Northrop Grumman B-2

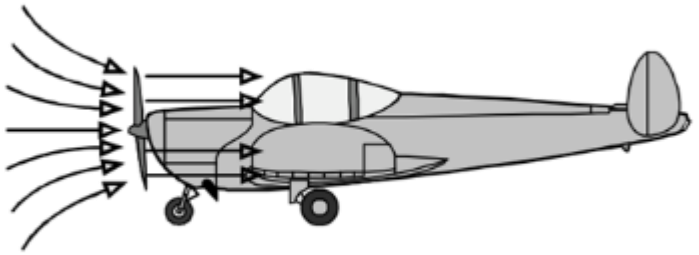
Source: Nicolai (2010)

Propulsion (1)

There a two main ways of generating thrust.

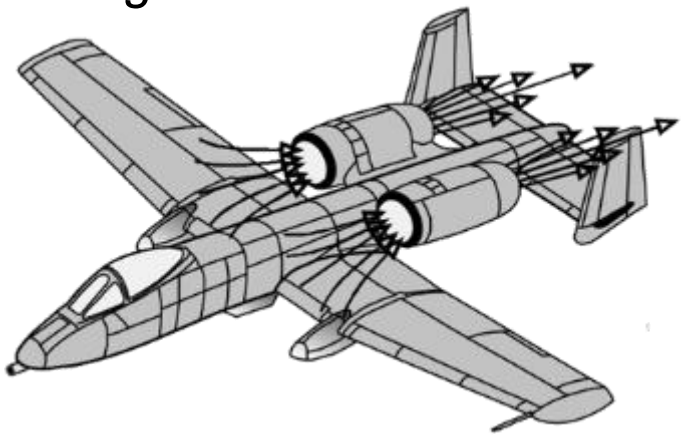
Mechanical Acceleration of Fluid

> Propeller Engine



Thermal Acceleration of Fluid

> Jet Engine

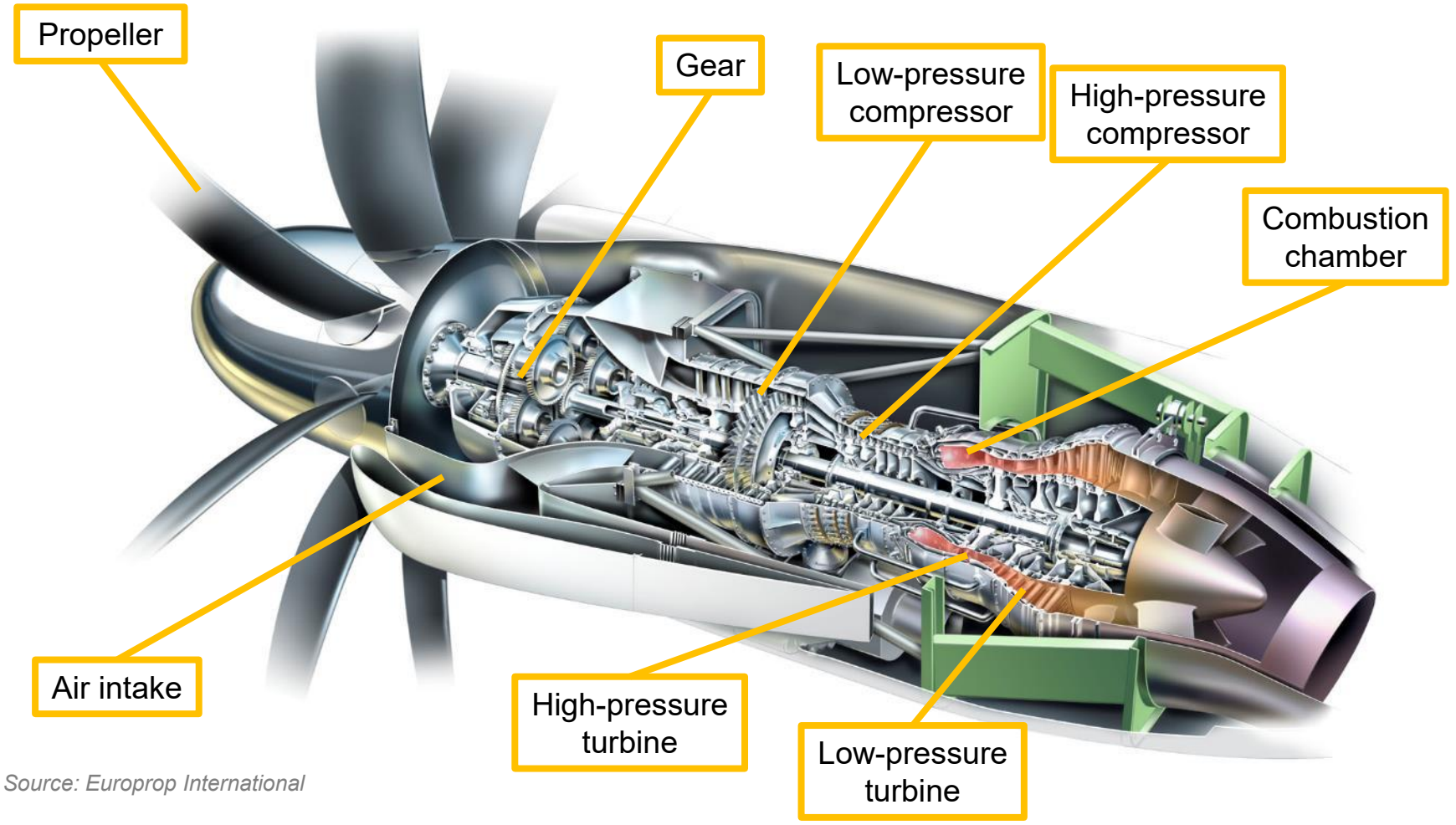


$$T = \dot{m}\,(v_{exit} - v_{in}) + A_{exit}\,(p_{exit} - p_0) - A_{in}\,(p_{in} - p_0)$$

Source: Bräunling (2009)

Propulsion (2)

Components of a modern turboprop engine.



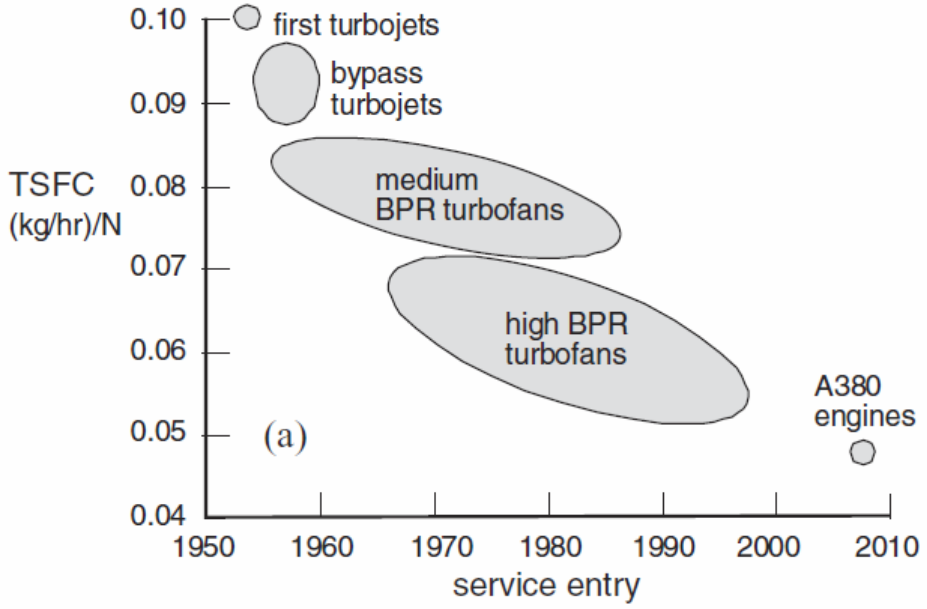
Source: Europrop International

Propulsion (4)

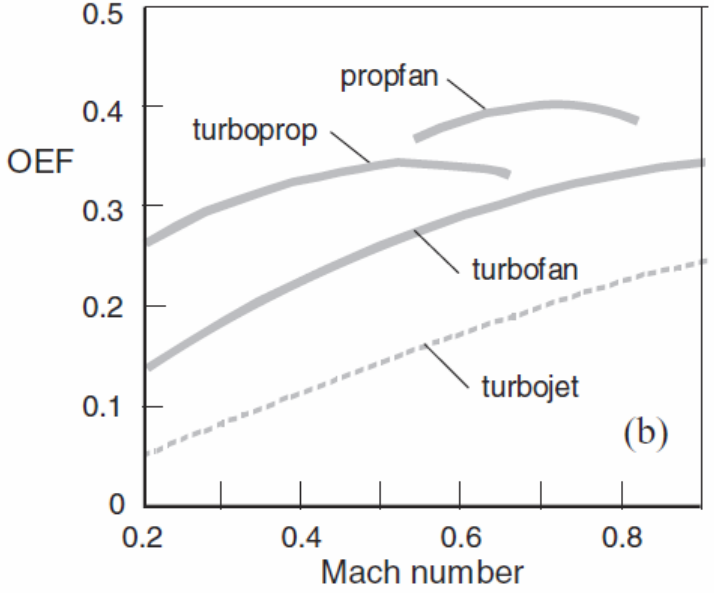
Thrust-specific fuel consumption (TSFC) and overall efficiency (OEF)

The selection of engine type strongly depends on the intended **mission profile** of the aircraft.

Evolution of engine efficiency



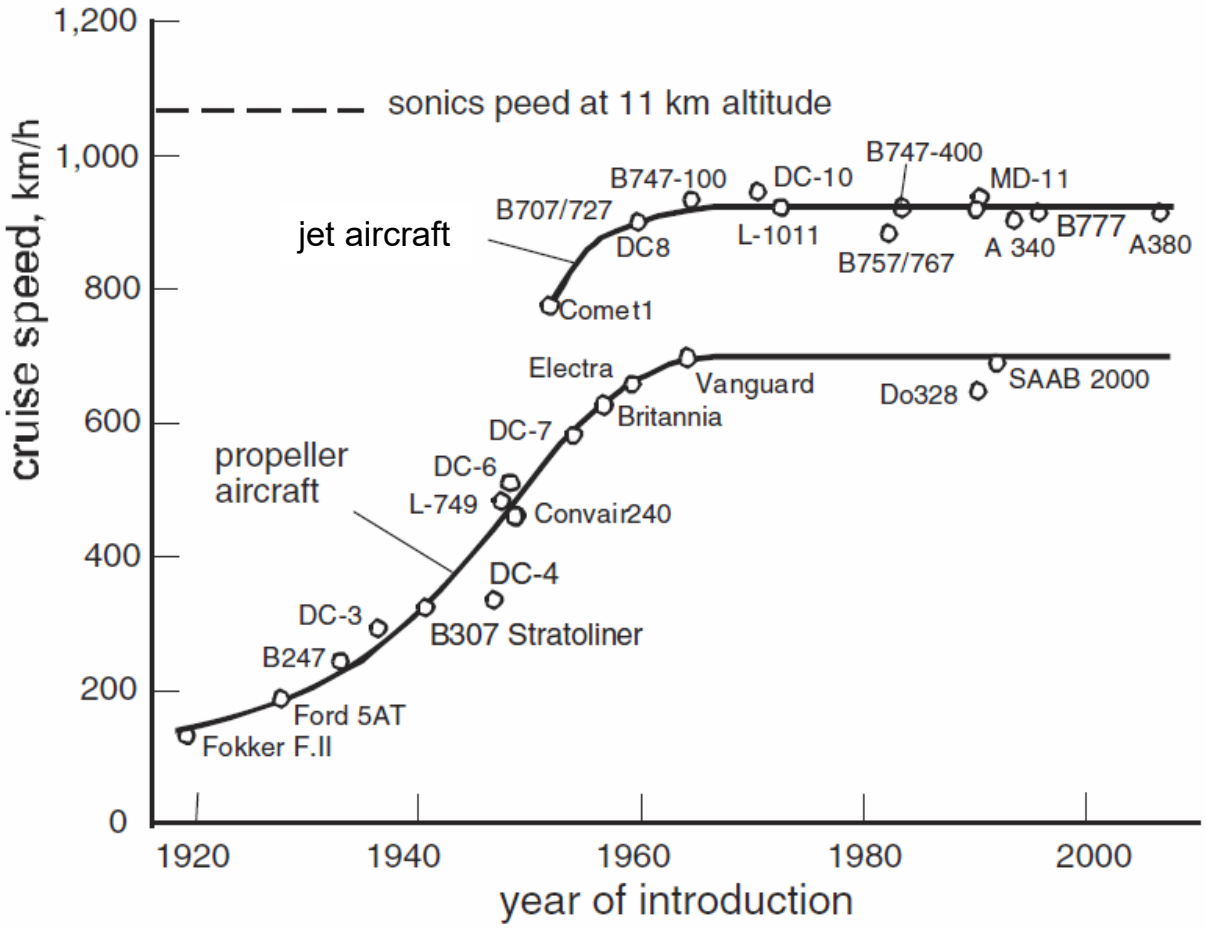
Selection of engine type according to cruise speed



Source: Torenbeek (2013)

Propulsion (5)

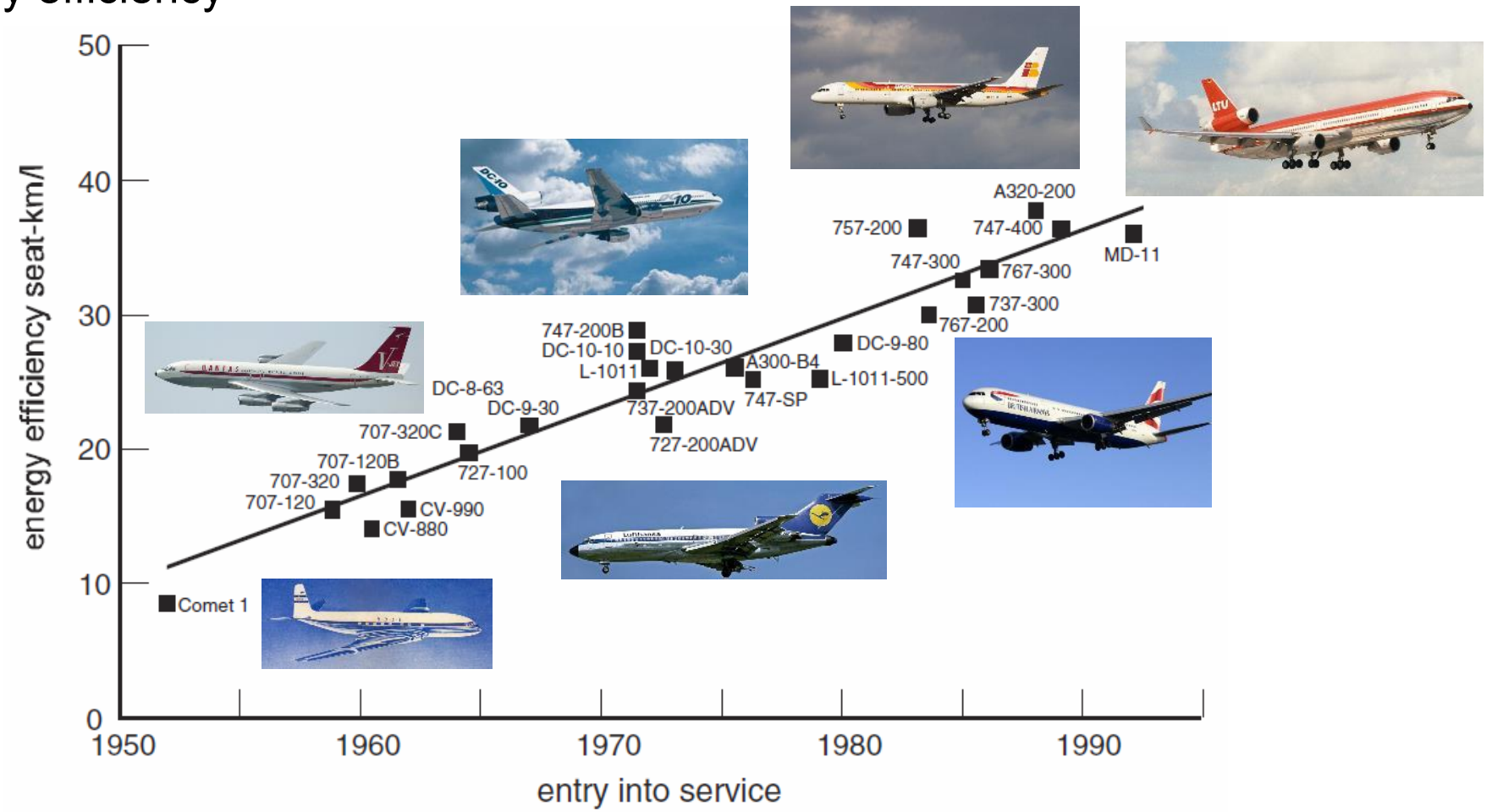
Historical Development of Cruise Speeds



Source: Torenbeek (2013)

Propulsion (6)

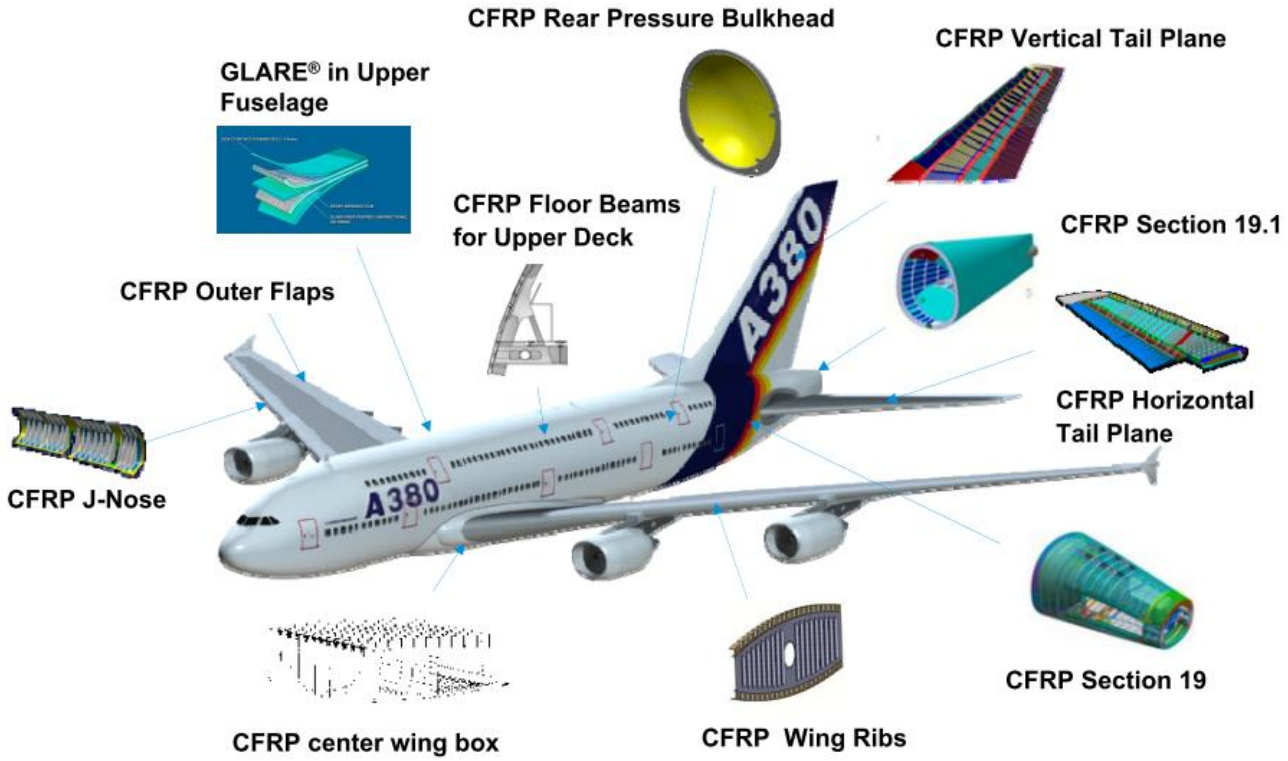
Historical energy efficiency



Source: Torenbeek (2013), NASA

Light-Weight Construction (1)

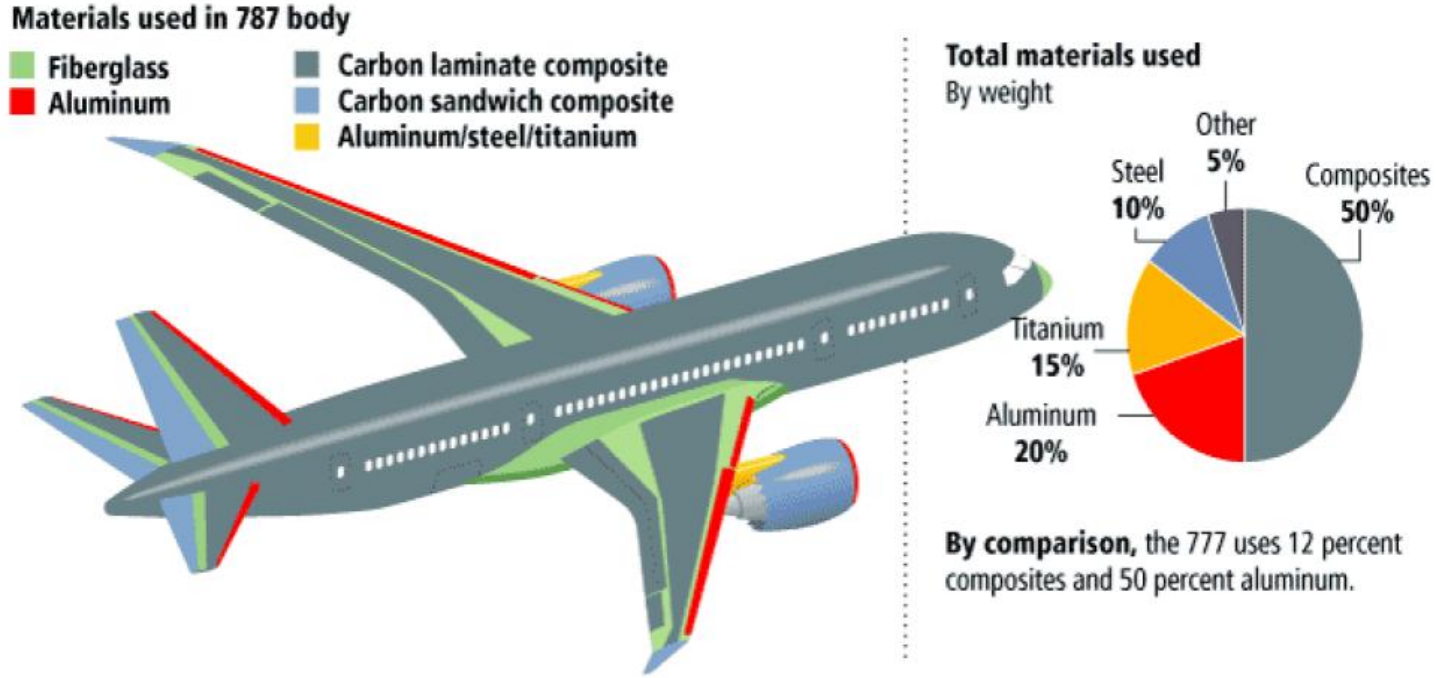
Status-Quo Technologies (Airbus A380)



Source: Stuttgart University (2013)

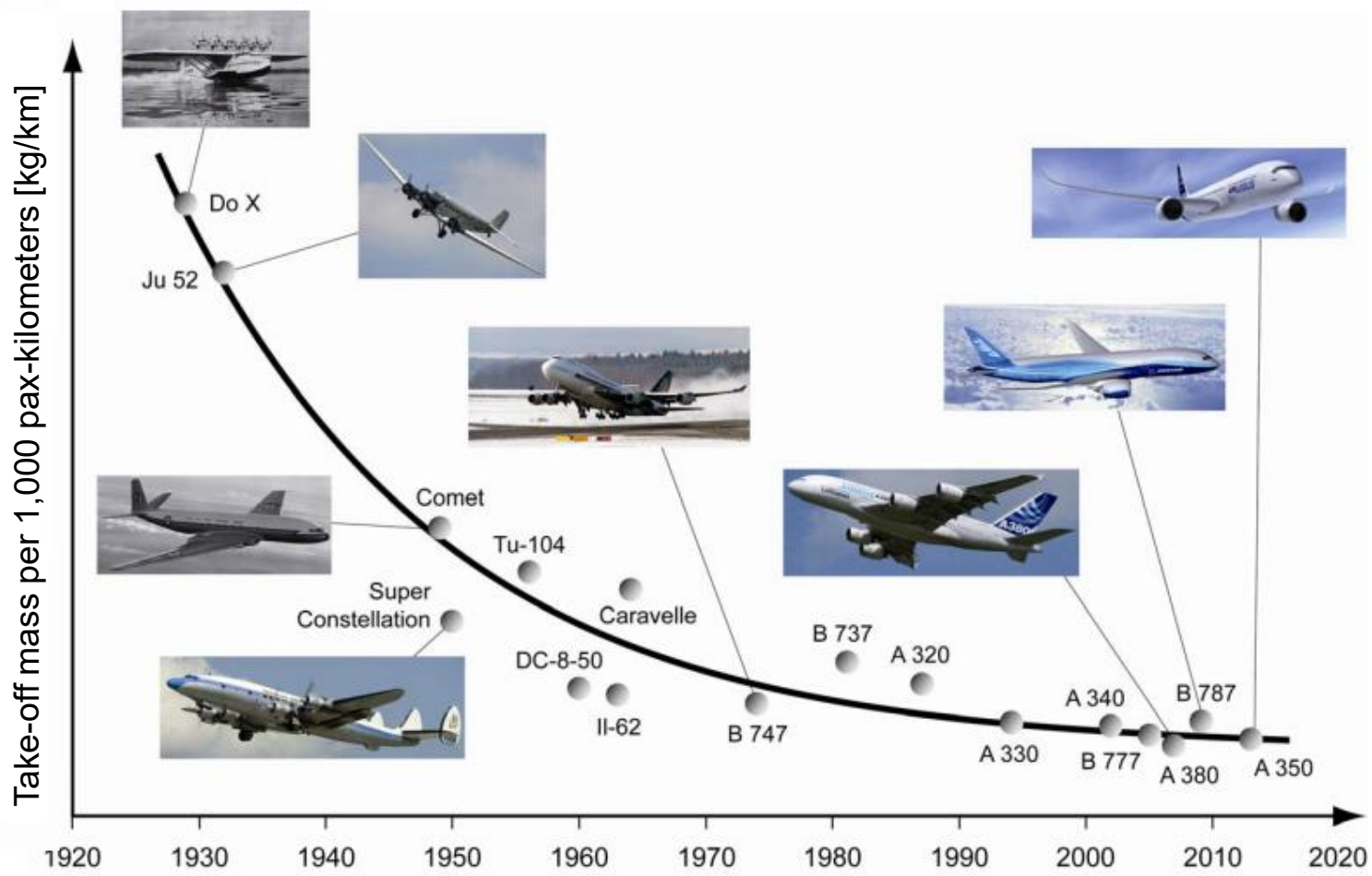
Light-Weight Construction (2)

Status-Quo Technologies (Boeing 787)



Source: Stuttgart University (2013)

Light-Weight Construction (3)



Source: Stuttgart University (2013)

Navigation

Goal: Navigate safely from departure to destination, **independent from visual flight conditions**

→ IFR (Instrument Flight Rules)

Navigation Aids

On-board Systems





On-ground Systems

Inertial Navigation

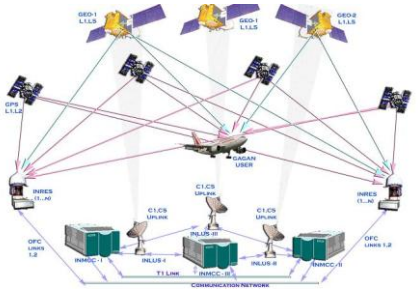
Radio Navigation





Radar Navigation

Satellite Navigation



Automation

Examples:



Automautic flight guidance (autopilot)

- Controls aircraft at most stages of flight



Flight management system (FMS)

- Calculates routing options, fuel consumption, optimum cruise altitude and speed, etc.

Full authority digital engine control (FADEC)

- Controls the engine performance in order to operate the engine at optimum efficiency in each flight stage



1. History of Civil Air Transport

1.3 Commercialization of Air Transport



Commercialization of Air Transport (1)

Civil air transport has developed from an elite way of travel to a **mass transportation system**.

1958



Luxury travel on board a DeHavilland Comet from London to New York.

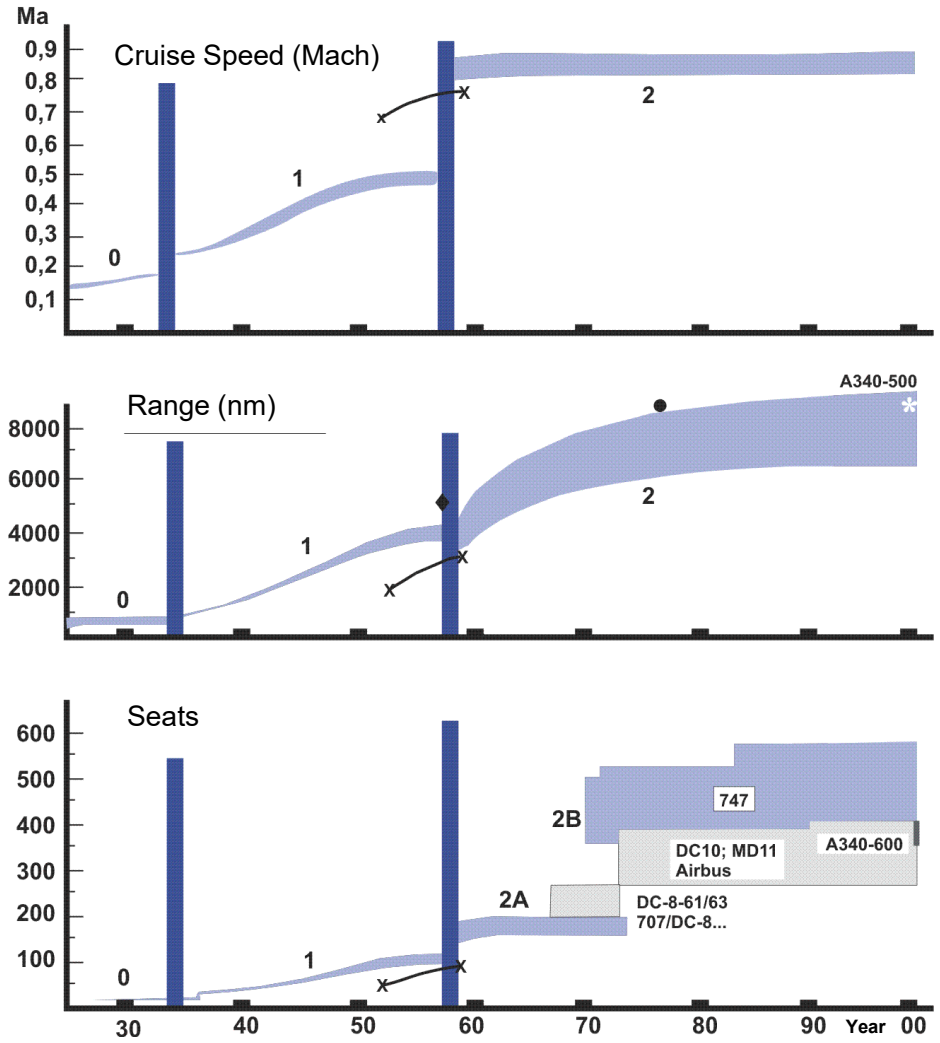
2010



Modern cabin layout of a European low-cost carrier.

Source: British Airways

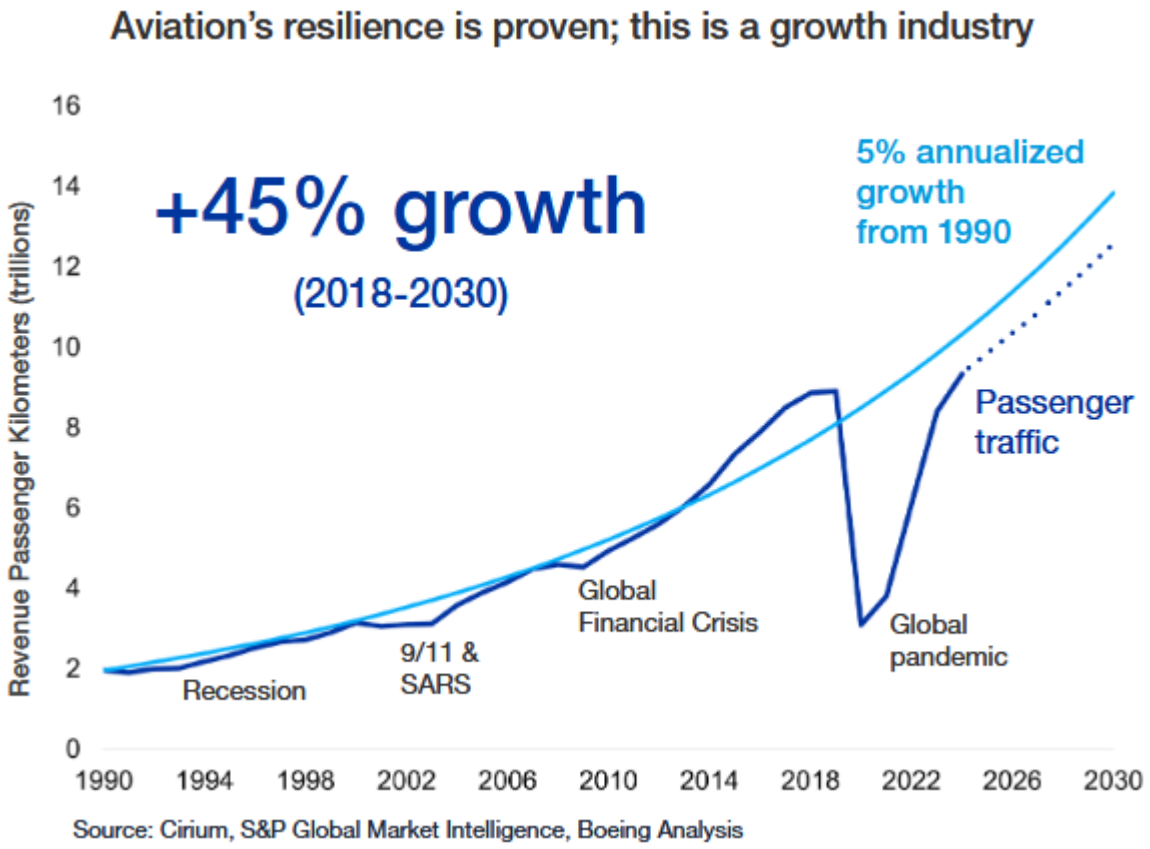
Commercialization of Air Transport (2)



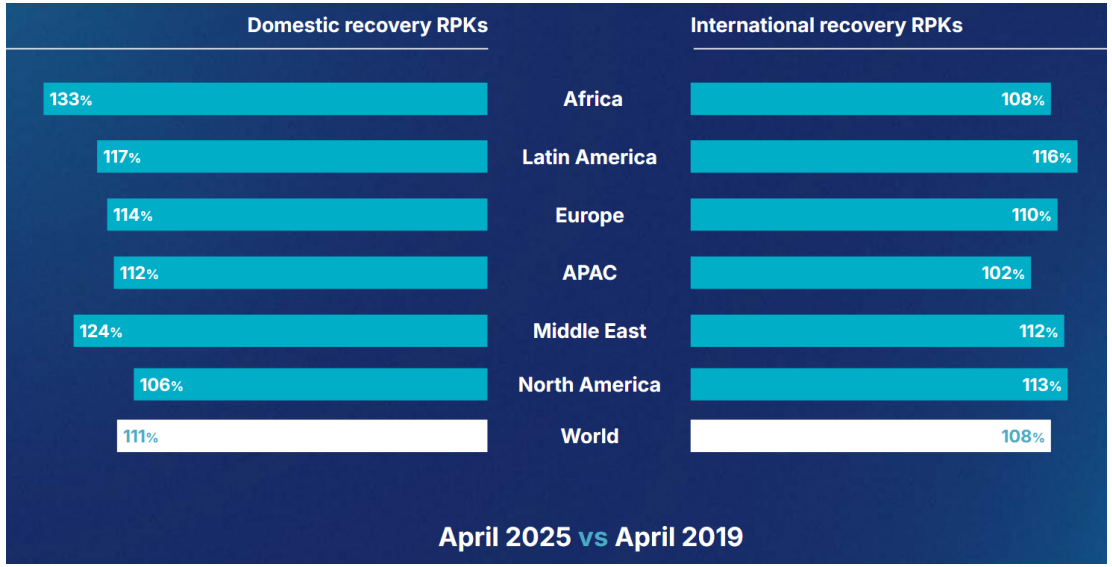
0 Piston Engine (< 1933)
1 Piston Engine (> 1933)
2 Jet Engines:
- 2A Narrow Body
- 2B Wide Body

Commercialization of Air Transport (3)

Major trends



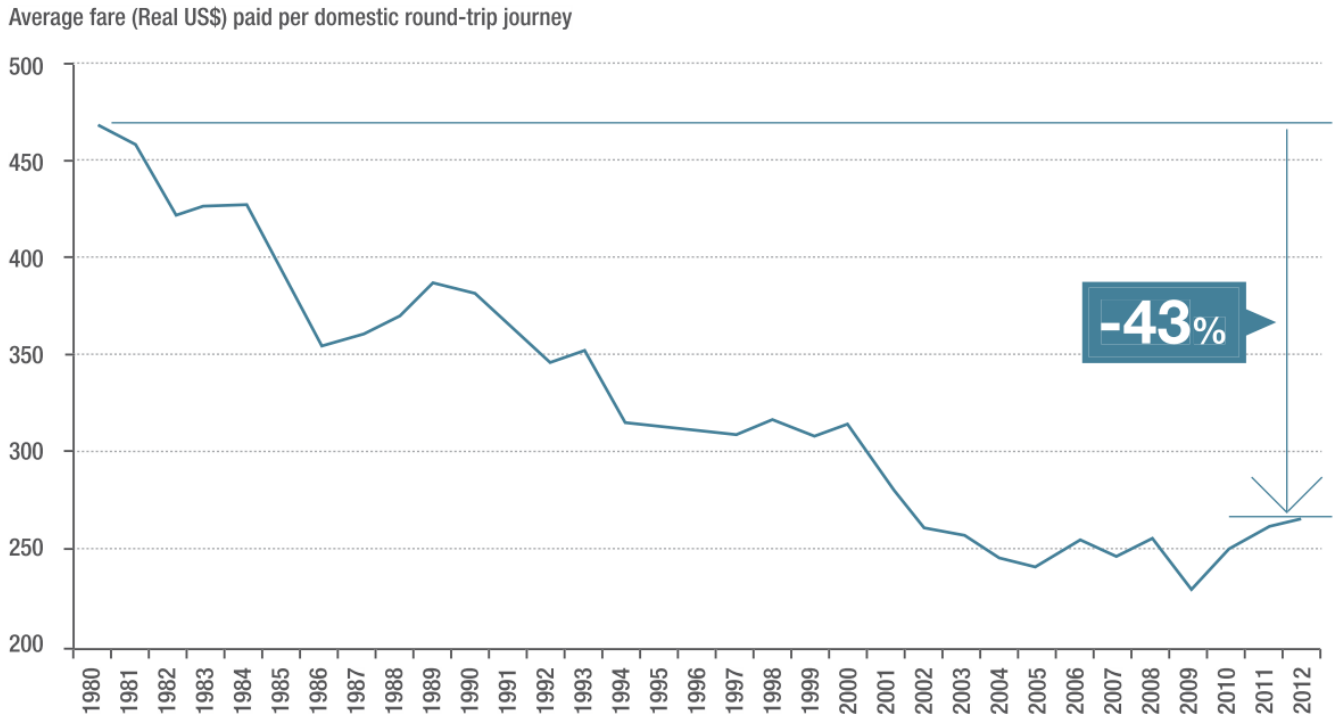
Airlines in all regions are at or above 2019 levels



Source: Airbus GMF 2025

Commercialization of Air Transport (4)

Major trends



Source: Airbus GMF 2013